

Project Development Phase

Code-Layout, Readability and Reusability

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

The Voice-Based Concept Understanding Analyser (VBCUA) is developed using a modular and maintainable software architecture. The project follows good programming practices by separating the application into independent modules responsible for speech transcription, semantic analysis, audio feature extraction, evaluation, visualization, and report generation. This modular design improves readability, simplifies debugging, and enables future enhancements Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project.	Yes	Improves code readability.
2	Proper File Structure	Source code organized into separate folders for models, utilities, reports and application files.	Yes	Easy to navigate and maintain.
3	Meaningful Variable Names	Variables such as audio_file, transcript, semantic_score, pause_ratio, and overall_score clearly indicate their purpose.	Yes	Enhances code understanding.
4	Meaningful Variable Names	Variables such as audio_file, transcript, semantic_score, pause_ratio, and overall_score clearly indicate their purpose	Yes	improves understanding.
5	Code Comments	Important preprocessing steps and model logic are documented using comments.	Yes	Easier maintenance
6	Modular Design	Data preprocessing, training, evaluation, and prediction are	Yes	Promotes code reuse.

		implemented in separate modules/functions.		
7	No Redundant Code	Duplicate code is avoided by using reusable functions.	Yes	Improves efficiency.
8	Error Handling	Input validation and exception handling are implemented for data loading and prediction.	Yes	Prevents runtime failures.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Audio Upload Module	Streamlit	Audio upload interface	High
2	Speech-to-Text Module	OpenAI Whisper	Audio transcription	High
3	Semantic Analysis Module	Sentence-BERT	Concept similarity evaluation	High
4	Audio Feature Extraction Module	Librosa	Pause ratio, RMS energy, filler word analysis	High
5	Visualization Module	Matplotlib	Waveform and performance charts	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Modular project with well-organized files
Readability	5	Meaningful variable names and clean formatting
Reusability	5	Independent reusable modules.
Documentation / Comments	5	Proper comments and descriptive functions.
Overall Score	5 / 5	Excellent code quality following software engineering best practices.